

Henry Yu

New York, NY | henry.yu@stonybrook.edu | 917-831-2139 | [linkedin.com/in/henryyu0505](https://www.linkedin.com/in/henryyu0505) | github.com/henhen0505

EDUCATION

Stony Brook University

Stony Brook, New York

Bachelor of Science in Computer Science, Focus in Algorithms and Systems

May 2027

Double major in Applied Mathematics and Statistics, Concentration in Probability and Optimization

Relevant Coursework: Data Structures, Algorithms, Analysis of Algorithms, Graph Theory, Systems Fundamentals, Computer Networks, Theory of Computation, Linear Algebra

WORK EXPERIENCE

BetterTherapy

New York, NY

Software Engineering Intern

May 2025 – August 2025

- Designed a **SQLAlchemy** data layer over **SQLite** for an automated LLM evaluation pipeline: schema with a reusable timestamp mixin, cascading request/response relationships, and a session decorator.
- Wrote atomic upserts (ON CONFLICT DO UPDATE) for a node blacklist, incrementing violation counts safely under concurrent writes.
- Avoided N+1 round trips with **selectinload** eager loading, added bulk inserts, and polled the **OpenAI Batch API** on 24-hour windows.

ReWild Long Island

Long Island, NY

Incoming Software Engineer, Dev Team

Fall 2026 – Summer 2027

Stony Brook University

Stony Brook, New York

Incoming Undergraduate Teaching Assistant

Fall 2026

CSE 260: Honors Data Structures & Programming Abstractions

PROJECTS

SEC Filing Evaluation Harness | Python, RAG, Anthropic Claude API, pandas, pytest

August 2026

- Built an evaluation harness for numerical accuracy on SEC filings against the open **LEDGER** benchmark, backed by an **80+** test suite in **pytest**.
- Generated **3,700+** questions on computed metrics (margins, ratios, YoY growth) from **SEC EDGAR XBRL** ground truth, extending the benchmark past extraction alone.
- Verified the benchmark against raw source data before building, identifying **six** discrepancies the published dataset omits.
- Ran a diagnostic evaluation over **25** questions across **~20** companies: **72%** of answers found the right figure at the wrong scale, masked by a **4%** raw accuracy score.

Playlist – Full-Stack Web App | React, Node.js, Express, MongoDB, PostgreSQL, Sequelize

October 2025

- Built a **MERN** app with **JWT** sessions and ownership checks on auth, playlist, and song **REST** endpoints.
- Built a database abstraction layer with **MongoDB** and **PostgreSQL** implementations behind one interface, switchable by environment flag with no controller changes.
- Added multi-field catalog search over the **YouTube Data API** and a **jsTPS** undo/redo stack for reversible edits, tested with **Vitest**.

Event-Driven Equity Research System | Python, FMP API, REST APIs

September 2025 – Present

- Designed an event-driven backend routing typed corporate events (earnings, ratings, FDA) to independent signal handlers, verified by **369** tests.
- Built an **FMP API** ingestion pipeline with retry logic and boundary schema validation, plus a CAR backtester across **1/5/10/20-day** horizons over **62** events.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Java, C/C++, Bash, HTML/CSS

Frameworks/Tools: React, Node.js, Express, MongoDB, PostgreSQL, SQLAlchemy, Git/GitHub, REST APIs, JWT Authentication, Linux, AWS, LLM APIs

Libraries: pandas, NumPy, statsmodels, matplotlib

Concepts: Data Structures & Algorithms, OOP, Full-Stack Web Development, System Design, API Design, Event-Driven Architecture, State Management, Testing (pytest)